The Fundamental Theorem of Calculus

Part 2: evaluating definite integrals

1 Evaluate each definite integral:

a $\int_1^3 (2x+1)\,dx$

b $\int_0^\pi \sin x\,dx$

c $\int_1^4 \sqrt{x}\,dx$

2 Evaluate $\int_0^1 e^{2x}\,dx$, to three decimal places.

Part 1: differentiating an integral

3 Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx}\int_2^x (t^3+1)\,dt$.

4 Find $\frac{d}{dx}\int_1^{x^2} \sin(t)\,dt$ using the chain rule with FTC.

5 Let $g(x) = \int_0^x \sqrt{4+t^2}\,dt$. Find $g'(3)$.

Average value and applications

6 Find the average value of $f(x) = x^2$ on $[0, 3]$.

7 A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \le t \le 4$. Find the total distance traveled, and the average velocity over this interval.

Further FTC practice

8 Evaluate $\int_0^2 (x^3 - 2x + 3)\,dx$.

9 Find $\frac{d}{dx}\int_x^5 \cos(t^2)\,dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.

10 Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.

Mean Value Theorem for integrals

11 Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).

12 Evaluate $\int_2^5 f(x)\,dx$ given that the average value of f on $[2, 5]$ is 7.

Integrals of piecewise functions

13 Evaluate $\int_0^4 f(x)\,dx$ where $f(x) = x$ for $x \leq 2$ and $f(x) = 2x - 2$ for $x > 2$.